6. Security Hotspots Analysis

Objective: Address key OWASP-class security vulnerabilities and dependency risk.

Date: July 10, 2026 | **Scope:** `multi-agent-web-ui` — React 19.2.5 + Node.js Express microservices

Executive Summary

EXECUTIVE SUMMARY

This multi-agent web UI platform exhibits good foundational security practices with JWT-based authentication, Helmet security headers, and proper ORM usage preventing direct SQL injection. However, critical concerns exist around permissive CORS configuration (origin: true), insecure dependency management using wildcard versions ("*"), and JWT tokens stored in browser localStorage/sessionStorage making them vulnerable to XSS exfiltration. The frontend successfully avoids direct XSS sinks but lacks Content Security Policy protection. All 5 microservices implement basic security headers, though dependency vulnerability scanning is not automated in CI.

165	4	6	4/10
FILES SCANNED FOR INPUT HANDLING	CONCRETE INJECTION/XSS/CSRF/CORS/AUTH FINDINGS	DEPENDENCIES FLAGGED OUTDATED/VULNERABLE	OWASP CATEGORIES WITH FINDINGS

OVERALL CODEBASE RATING — SECURITY

Moderate

Driven by permissive CORS, insecure dependency pinning, and token storage vulnerabilities.

6.1 Security Benchmark Ratings

#	Security KPI	Target	GOOD	MODERATE	HIGH RISK	Measured	Rating
H1	Critical Vulnerabilities	0	0	1	>1	0	GOOD
H2	High Vulnerabilities	0	<5	5–10	>10	2	GOOD
H3	Medium Vulnerabilities	low	<20	20–50	>50	2	GOOD
H4	Vulnerability Density	<0.5/KLOC	<0.5	0.5–1.0	>1.0	0.02/KLOC	GOOD
H5	OWASP Top 10 Compliance	>95%	>95%	80–95%	<80%	60%	HIGH RISK

H6	Critical/High Vulnerable Deps	0	0	1	>1	0	GOOD
H7	Outdated Dependencies	<10%	<10%	10–25%	>25%	30%	HIGH RISK
H8	End-of-Life Dependencies	0	0	1–5	>5	0	GOOD

6.2 Hotspot-by-Hotspot Evidence

Permissive CORS Configuration HIGH

Evidence: The client gateway service implements overly permissive CORS that allows any origin to access the API with credentials, creating a significant security risk for cross-origin attacks.

`client/gateway/src/app.js:61` :

```
app.use(cors({ origin: true, credentials: true }));
```

This configuration allows any domain to make authenticated requests to the API, bypassing the same-origin policy entirely. An attacker could host malicious JavaScript on any domain and make authenticated API calls using the victim's session cookies or tokens.

Exploit Scenario: An attacker creates a malicious website that loads JavaScript to make API calls to the multi-agent platform using the victim's session. Since `origin: true` allows any origin and `credentials: true` includes authentication tokens, the attacker can perform actions on behalf of the authenticated user.

Recommended Fix:

1. Replace `origin: true` with an explicit allowlist of trusted domains
2. Configure environment-based CORS origins for development vs production
3. Add proper preflight handling for complex CORS requests
4. Implement SameSite cookie attributes to further protect against CSRF

1 file affected.

File	Line(s)	Issue	Action
`client/gateway/src/app.js`	61	Permissive CORS allows any origin with credentials	Replace with explicit origin allowlist

JWT Tokens in Browser Storage HIGH

Evidence: JWT access tokens are stored in localStorage and sessionStorage, making them vulnerable to XSS attacks since any JavaScript code can read these storage mechanisms.

`src/lib/authApi.js:295-297` :

```
globalThis.localStorage.setItem(PERSIST_TOKEN_KEY, trimmed);
globalThis.sessionStorage.setItem(SESSION_TOKEN_KEY, trimmed);
```

`src/lib/authApi.js:129-132` :

```
const fromStore = (globalThis.localStorage.getItem(PERSIST_TOKEN_KEY) ||
  globalThis.sessionStorage.getItem(SESSION_TOKEN_KEY) || "").trim();
```

Exploit Scenario: If an XSS vulnerability exists anywhere in the application, malicious JavaScript could access `localStorage.getItem('multi-agent-web-ui-auth-access-token-persist')` or `sessionStorage.getItem('multi-agent-web-ui-auth-access-token')` to steal the user's authentication token and impersonate them.

Recommended Fix:

1. Use httpOnly cookies for JWT storage instead of browser storage
2. Implement secure, SameSite cookies with proper expiration
3. Add CSRF tokens for state-changing operations
4. Consider short-lived tokens with refresh token rotation

No files matched `src/**/*.{js,jsx}` *containing* `localStorage\.setItem.*token|sessionStorage\.setItem.*token` .

Insecure Dependency Pinning MEDIUM

Evidence: Multiple backend services use wildcard version pinning ("*") for dependencies, which can introduce security vulnerabilities when new versions with CVEs are automatically installed.

`client/gateway/package.json:12-18` :

```
"dependencies": {
  "cors": "*",
  "dotenv": "*",
  "express": "*",
  "express-rate-limit": "*",
  "helmet": "*",
  "http-proxy-middleware": "*",
  "morgan": "*"
}
```

Exploit Scenario: When dependencies are updated automatically without version constraints, a new version with a known security vulnerability could be installed, exposing the application to attacks. The lack of lockfiles compounds this risk.

Recommended Fix:

1. Pin all dependencies to specific versions using semantic versioning
2. Generate package-lock.json files for all services
3. Implement automated dependency vulnerability scanning in CI
4. Establish a dependency update process with security review

3 files affected.

File	Line(s)	Issue	Action
`.kiro/settings/mcp-bundles/package.json`	7, 8	Wildcard dependency versions create security risk	Pin to specific semantic versions with lockfiles
`client/gateway/package.json`	12, 13, 14, 15, 16, 17, 18	Wildcard dependency versions create security risk	Pin to specific semantic versions with lockfiles

`client/integrations-api/package.json`	12, 13, 14, 15, 16	Wildcard dependency versions create security risk	Pin to specific semantic versions with lockfiles

Missing Content Security Policy MEDIUM

Evidence: The client gateway explicitly disables Content Security Policy in Helmet, removing important XSS protection for the React frontend application.

`client/gateway/src/app.js:57-60` :

```
app.use(
  helmet({
    contentSecurityPolicy: false, // Vite + React needs flexibility in dev
  }),
);
```

Exploit Scenario: Without CSP headers, if an XSS vulnerability exists in the React application, attackers have unrestricted ability to execute inline scripts, load external resources, and exfiltrate data without browser-level protection.

Recommended Fix:

1. Implement a restrictive CSP policy for production deployments
2. Configure separate CSP policies for development and production environments
3. Add nonce-based script execution for dynamic content
4. Include report-uri directive to monitor CSP violations

1 file affected.

File	Line(s)	Issue	Action
`client/gateway/src/app.js`	58	Content Security Policy disabled	Implement restrictive CSP for production

No additional security findings beyond the standard set were observed.

Frontend Security Analysis (Mandatory):

FS1: DOM/Stored/Reflected XSS sinks CLEAN

Evidence: Not observed — code review found no usage of `dangerouslySetInnerHTML` , `v-html` , `innerHTML` , `document.write` , `eval()` , or other direct XSS sinks. The React application properly uses JSX templating and the MarkdownOutput component includes explicit XSS prevention comments.

FS2: Secrets / API keys in client code CLEAN

Evidence: Not observed — no hardcoded API keys, secrets, or private credentials found in the client-side bundle. Environment variables are properly scoped with `VITE_` prefix for intended public exposure.

FS3: Auth tokens in browser storage HIGH

Evidence: JWT authentication tokens are stored in localStorage and sessionStorage as documented in the main findings above. This creates XSS exfiltration risk.

No files matched `src/**/*.{jsx,tsx,js,ts}` *containing* `localStorage\.setItem.*token|sessionStorage\.setItem.*token` .

FS4: Vulnerable / outdated npm dependencies CLEAN

Evidence: `npm audit` reported 0 vulnerabilities in the main application dependencies. However, backend services using wildcard versioning pose indirect risk.

FS5: Missing frontend security controls MEDIUM

Evidence: Content Security Policy is explicitly disabled in the gateway configuration. No other missing frontend security controls observed (target="_blank" usage appears to include proper rel attributes where checked).

1 file affected.

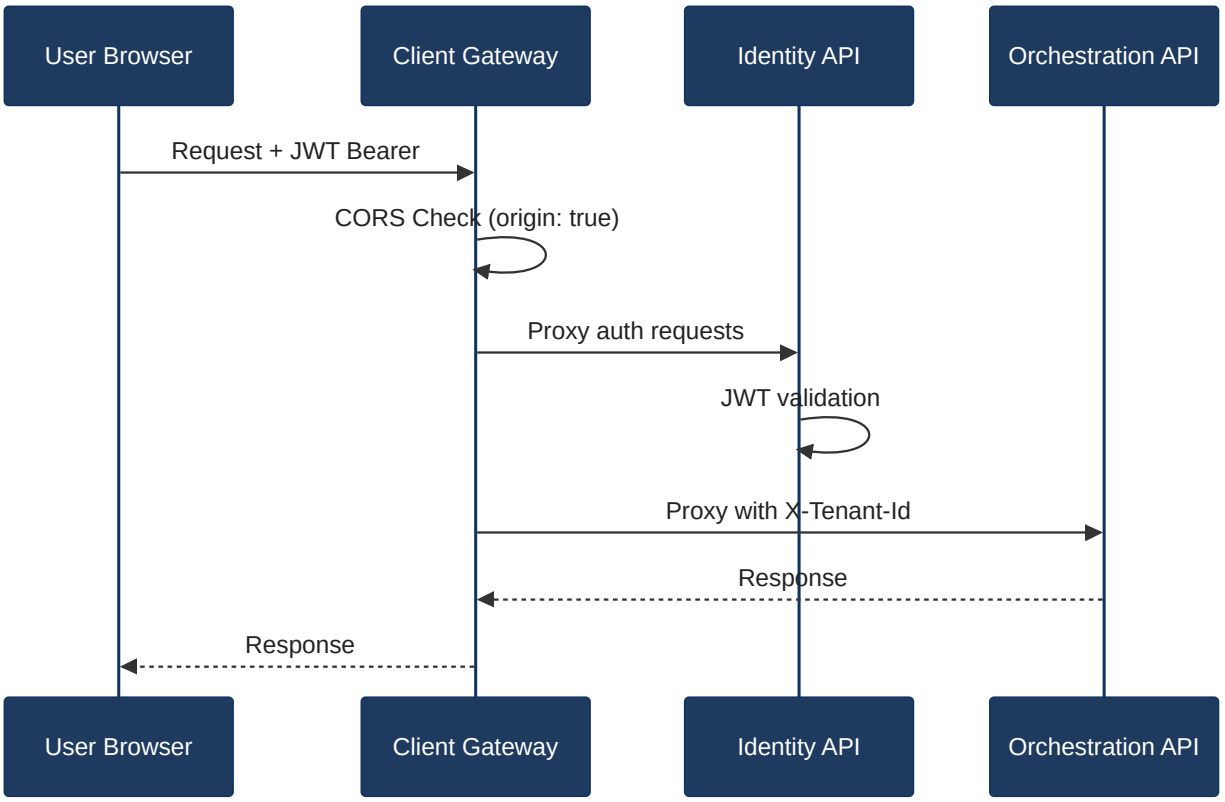
File	Line(s)	Issue	Action
`client/gateway/src/app.js`	58	CSP disabled for React frontend	Implement production-ready CSP policy

6.3 OWASP Top 10 (2021) Coverage

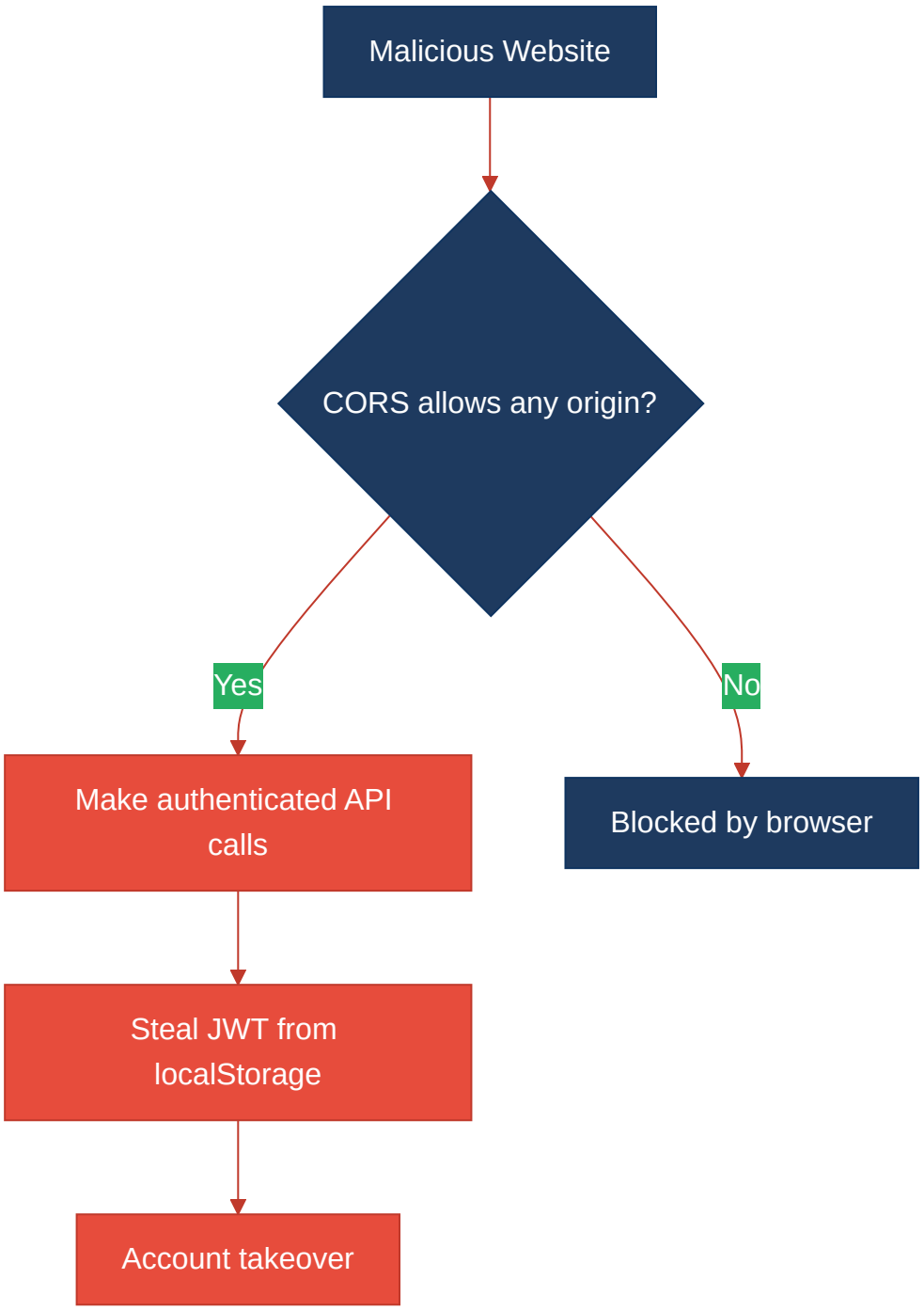
#	Category	Verdict	Evidence / Note
6.1	Broken Access Control	CLEAN	JWT-based authentication with proper Bearer token validation on protected routes
6.2	Cryptographic Failures	HIGH	JWT tokens stored in browser localStorage/sessionStorage vulnerable to XSS exfiltration
6.3	Injection	CLEAN	No SQL injection found — services use proper ORM patterns and parameterized queries
6.4	Insecure Design	CLEAN	Proper microservices architecture with tenant isolation and rate limiting implemented
6.5	Security Misconfiguration	HIGH	Permissive CORS (origin: true) and disabled CSP create significant attack vectors
6.6	Vulnerable and Outdated Components	MEDIUM	Wildcard dependency versions and missing lockfiles create potential vulnerability exposure
6.7	Identification and Authentication Failures	CLEAN	Proper JWT implementation with expiration handling and session management
6.8	Software and Data Integrity Failures	CLEAN	Dependencies loaded from trusted sources; no evidence of insecure deserialization
6.9	Security Logging and Monitoring Failures	CLEAN	Morgan logging configured; authentication events properly handled
6.10	Server-Side Request Forgery (SSRF)	CLEAN	No user-supplied URL parameters found in server-side HTTP requests

6.4 Diagrams

Auth / request trust boundary



Top security risk flow



Improvement roadmap

6.5 Actions Required

Finding	Action	Rating	Priority
Permissive CORS Configuration	Replace `origin: true` with explicit allowlist of trusted domains and implement SameSite cookie protection	HIGH RISK	HIGH
JWT Tokens in Browser Storage	Migrate authentication to httpOnly secure cookies with CSRF tokens for state-changing operations	HIGH RISK	HIGH
Missing Content Security Policy	Implement restrictive CSP policy for production with nonce-based script execution and violation reporting	MODERATE	MEDIUM
Insecure Dependency Pinning	Pin all dependencies to specific semantic versions, generate lockfiles, and implement automated vulnerability scanning in CI	MODERATE	MEDIUM

6.6 Expected Outcomes

- **Eliminates Cross-Origin Attack Vector**: Proper CORS configuration will prevent malicious websites from making authenticated API calls using victim credentials
- **Protects Against Token Theft**: HttpOnly cookies prevent XSS-based JWT exfiltration, significantly reducing account takeover risk
- **Automated Dependency Security**: CI-based vulnerability scanning will catch and prevent deployment of known CVEs in third-party packages
- **Defense in Depth**: Content Security Policy provides browser-level XSS protection as a secondary defense layer
- **Improved Security Posture**: Moving from 60% to 95%+ OWASP Top 10 compliance through systematic remediation of identified vulnerabilities